MIL-S-55389(EL)
AMENDMENT 2
21 April 1978
SUPERSEDING
AMENDMENT 1
9 April 1975

MILITARY SPECIFICATION

SUPPRESSOR, ELECTRICAL TRANSIENT, MX-7778A/GRC

This amendment forms a part of Military Specification MIL-S-55389(EL), dated 30 March 1973.

PAGE 1

* 2.1, List of FEDERAL SPECIFICATIONS: Delete and substitute:

"FEDERAL

T-T-871	Twine, Cotton, Wrapping
NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, and Seals
PPP-B-20	Bag, Cotton, Mailing
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated Triplewall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, Nonmetallic, (And Connectors)
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced"

FSC 5915

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* 2.1, List of MILITARY SPECIFICATIONS: Delete and substitute:

"MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-B-117	Bags, Sleeves and Tubing-Interior Packaging
MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and associated Electrical Equipment
MIL-V-173	Varnish, Moisture-And-Fungus-Resistant (for Electronic And Associated Equipment)
MIL-P-11268	Parts, Materials, And Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, For Packaging"

* 2.1, List of MILITARY STANDARDS: Delete and substitute:

"MILITARY

MIL-STD-105	Sampling, Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-252	Classification of Visual and Mechanical Defects For Equipment, Electronic, Wired, and Other devices
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities"

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* 3.4.2, Delete and substitute:

"3.4.2 Voltage Limitation. With the output of the transient suppressor open circuited and the input voltage adjusted such that the zener diode draws 1 ± 0.1 amperes at two (2) seconds of this voltage application, the maximum allowable output voltage shall be within a range of 33 to 36 volts referred to a zener case temperature of 86 ± 5.5 degrees F (subtract 17.2 mV per degree F from the output voltage for a lower zener case temperature and add 17.2 mV per degree F to the output voltage for a higher zener case temperature). (See 4.7.1)"

PAGE 6

* 3.5.7, line 2: Delete "be capable of being immersed" and substitute "withstand immersion".

PAGE 7

* 3.7.1, line 2: Delete "X24087" and substitute "24087"

PAGE 9

* 4.2, Delete and substitute:

"4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include Packaging.

b. Inspection covered by subsidiary documents (See 4.4)

c. Quality Conformance inspection:

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Quality conformance inspection of Packaging. (See 4.11)

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* Table I: Delete and substitute:

" TABLE I FIRST ARTICLE INSPECTION

INSPECTIONS	REQ PARA.	TEST PARA.	UNIT 1	UNIT 2	UNIT 3
Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units		
Bounce, Preconditioning	3.3	4.5.4	Inspections to be performed on all units		
GROUP A Inspection	(See Table II)		Inspection to be performed on all units		
GROUP B Inspection	(See Table III)		Inspection to be performed on all units		
Environmental					
Low Temperature	3.5.1.2	4.6.1.2	1		
High Temperature	3.5.1.1	4.6.1.1	2		
Orientation	3.5.8	4.6.8	3		
Humidity(Moisture Resistance)	3.5.2	4.6.2	4		
Altitude	3.5.3	4.6.3		1	
Salt Fog	3.5.9	4.6.9		2	
Immersion	3.5.7	4.6.7			1
Sand and Dust	3.5.6	4.6.6			2
Vibration, Resonance and Vibration Cycling	3.5.4.1	4.6.4.1			3
Vibration, Bounce	3.5.4.2	4.6.4.2			4
Shock	3.5.5	4.6.5			5
Fungus	3.5.10	4.6.10			6
Pulse Circuit	3.4.4.1	4.7.3.1		Test Equipment	"

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* Table III: Add as last inspection:

"INSPECTION	REQ PARA	INSP PARA	AQL
Immersion	3.5.7	4.6.7	1.0%

NOTE: Special Inspection Level is S4 for each Group B inspection specified."

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* Table IV: Delete and substitute:

"TABLE IV - GROUP C INSPECTION

INSPECTION	REQ. PARA.	INSP. PARA
SUBGROUP IA		
High Temperature	3.5.1.1	4.6.1.1
Low Temperature	3.5.1.2	4.6.1.2
Altitude	3.5.3	4.6.3
SUBGROUP IB		
Humidity (Moisture Resistance)	3.5.2	4.6.2
SUBGROUP IC		
Vibration, Resonance and Vibration Cycling	3.5.4.1	4.6.4.1
Vibration, Bounce	3.5.4.2	4.6.4.2
Shock	3.5.5	4.6.5
SUBGROUP IIA		
Fungus	3.5.10	4.6.10 (See Note 1)
Sand and Dust	3.5.6	4.6.6
SUBGROUP IIB		
Orientation	3.5.8	4.6.8
Salt Fog	3.5.9	4.6.9
SUBGROUP III		
Pulse Circuit Test	3.4.4.1	4.7.3.1

NOTE 1: This test shall be performed on a separate sample."

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* 4.5.3.1.1: Delete and substitute:

"4.5.3.1.1 Subgroups IA, 1B, 1C, Table IV. - Two units shall be selected at random from each 3000 equipments produced, or 2 every 6 months if less than 3000 units are produced in a 6 month period, and subjected to each inspection specified. The first sample shall be selected at the start of the contract from the First Quality Conformance Inspection Lot."

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* 4.5.3.1.2: Delete and substitute:

"4.5.3.1.2 Subgroups IIA and IIB, Table IV. - Two units shall be selected at the start of the contract from the First Quality Conformance Inspection Production Lot and 2 units from the first 50 units following the midpoint production quantity shall be selected. No subsequent Group C inspection for the tests in these subgroups will be required."

* 4.5.3.1.3: Delete and substitute:

"4.5.3.1.3 Subgroup III, Table IV. - The pulse circuit test shall be performed once for each 1000 equipments produced."

* 4.5.3.1.3.1: Delete in its entirety.

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* 4.10: Delete and substitute:

"4.10 Rough handling test (Packaging). - When the rough handling test in accordance with MIL-P-116 is required by contract (See 6.2(j)), the unit shall be subjected to the tests of 4.7.1 thru 4.7.7 to determine freedom from malfunction caused by the rough handling."

* 4.11: Delete and substitute:

"4.11 Quality Conformance inspection of Packaging. - Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5."

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* 5. PREPARATION FOR DELIVERY: Delete and substitute:

"PACKAGING

5.1 Preservation. Preservation shall be level A, B or commercial, as specified (see 6.2(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Suppressor, Electrical Transient MX-7778A/GRC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Suppressor, Electrical Transient MX-7778A/GRC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods or submethods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed submethod IC-1.

5.1.1.4.2 Cable Assembly, CX-10613/G. Each cable assembly shall be unit packed method III as follows: coil each cable into a coil of proportionate dimensions and tie in three places with cotton twine conforming to T-T-871. Secure the connectors to the coil with the ties.

5.1.1.4.3 Template. Each template shall be unit packed submethod IC-1 as follows: place the template within a close-fitting bag conforming to MIL-B-117, type II, class C, style 1 and heat seal the closure.

5.1.1.4.4 Mounts and nuts. Items bearing the same stock number shall be unit packed method III by placing the items within a draw-string bag, conforming to PPP-B-20, close the bag with string ties.

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5.1.1.4.5 Suppressor, Electrical Transient MX-7778A/GRC. Each Suppressor, Electrical Transient MX-7778A/GRC shall be unit packed method III as follows: cushion the unit, on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned unit and the items, unit packed as specified in 5.1.1.4.2 through 5.1.1.4.4, within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2(b)). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Suppressor, Electrical Transient MX-7778A/GRC, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

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5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Suppressor, Electrical Transient MX-7778A/GRC unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Suppressor, Electrical Transient MX 7778A/GRC, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STL-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188."

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* 6.2(b): Delete and substitute:

"(b) Level A, level B or commercial preservation and packing (see section 5)."

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* 6.2(d): Delete and substitute:

"(d) First article inspection:

(1) Three sample units of each item cited in the section 1 are generally required so that lengthy and potentially damaging environmental tests can be performed on several sample units. (See 3.1)

(2) First article pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon."

* 6.2: Add:

"(j) When rough handling test is required.

(k) Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods."

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* 6.3: Delete and substitute:

"6.3 Color. The color chip furnished by the procuring agency (See 3.7.1) will be color chip No. 24087, and may be obtained from the Specifications Activity, Printed Materials Supply Division, Building 197, Naval Weapons Plant, Washington, D.C. 20407."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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